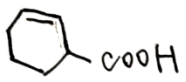
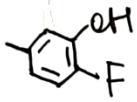
姓名：王梓遇

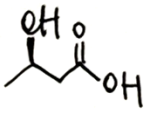
学号：2021521019

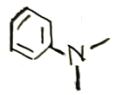
学院：生命科学与技术学院

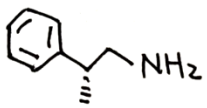
年级：2021

1. a) 

b) 

c) 

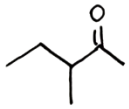
d) 

e) 

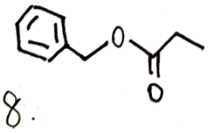
2. a) Aromatic: (b)、(c)、(e).

b) Acidity: (d) > (a) > (b) > (c)

3. a) $IDH = \frac{1}{2}(6\times2+2+2) = 1$

Structure: 

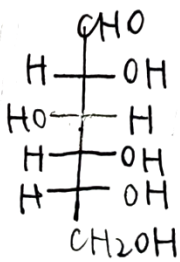
b) $IDH = \frac{1}{2}(10\times2+2-12) = 5$

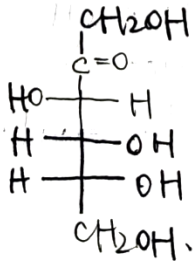
Structure: 

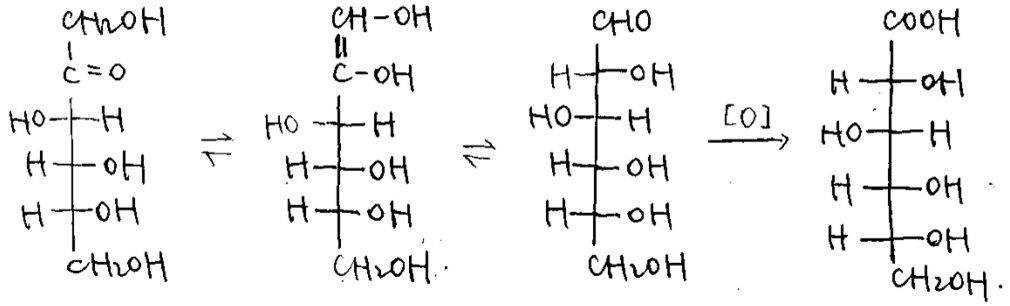
^{13}C peak: 8.

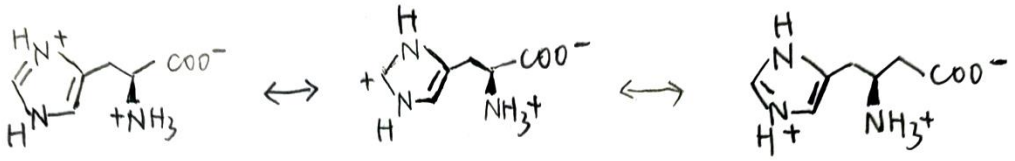
4. 1) A: α linkage, B: β linkage

2) 不是

3) A: 

B: 

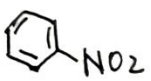
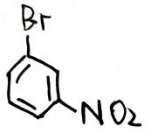
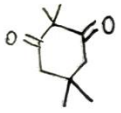
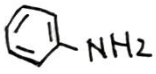
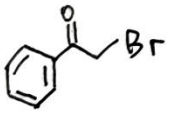
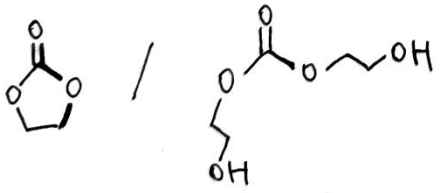
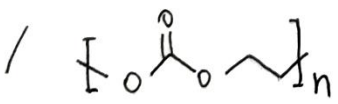
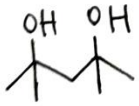
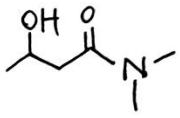
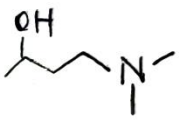
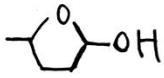
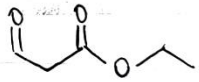
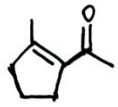
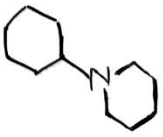
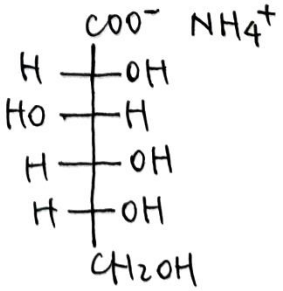
4). 

5. 1) 

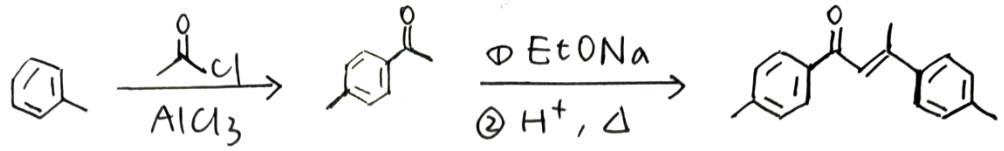
2) Imidazole group: 6.10 $-NH_2$: 9.18 $-COOH$: 1.77

3) $pI = \frac{1}{2}(pKa_1 + pKa_2) = \frac{1}{2}(9.18 + 6.10) = 7.64$

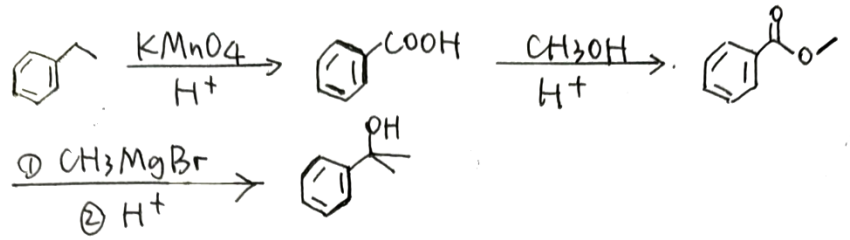
4) imidazole group: ⊕ $-NH_2$: ⊕ $-COOH$: ⊖
total net charge: +1

6. ① 
② 
③ 
④ 
⑤ 
⑥ 

⑦ 
⑧ 
⑨ 
⑩ 
⑪ 
⑫ 
⑬ 
⑭ 

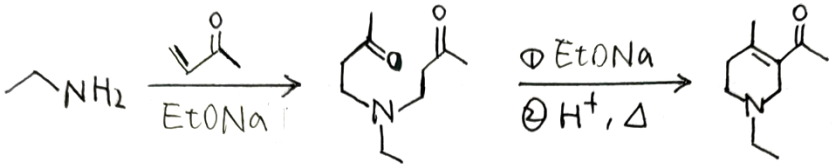
7. 1).



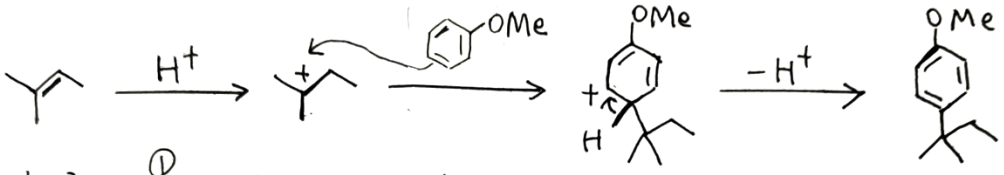
2).



3)



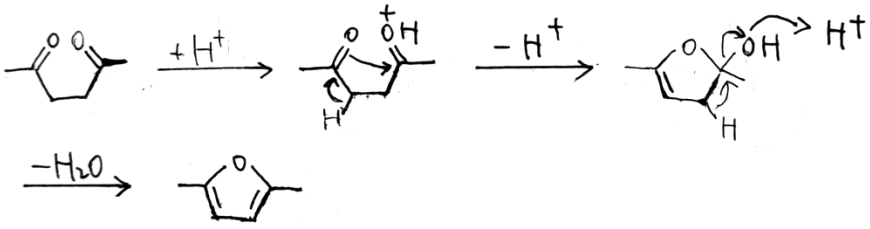
8. 1)



选择性：① 甲氧基是邻对位取代基，对位位阻小，易于反应；

② 该烯烃中叔碳生成的碳正离子较稳定

2).



3).

